

Configuring Color and Text with CSS

Chapter-4

What is CSS?

- Stands for Cascading Style Sheets
- Separates presentation from content
- Configures text, color, and layout

Advantages of Cascading Style Sheets

- Typography and page layout can be better controlled
- Style is separate from structure
- Styles can be Stored
- Documents are potentially smaller
- Site maintenance is easier

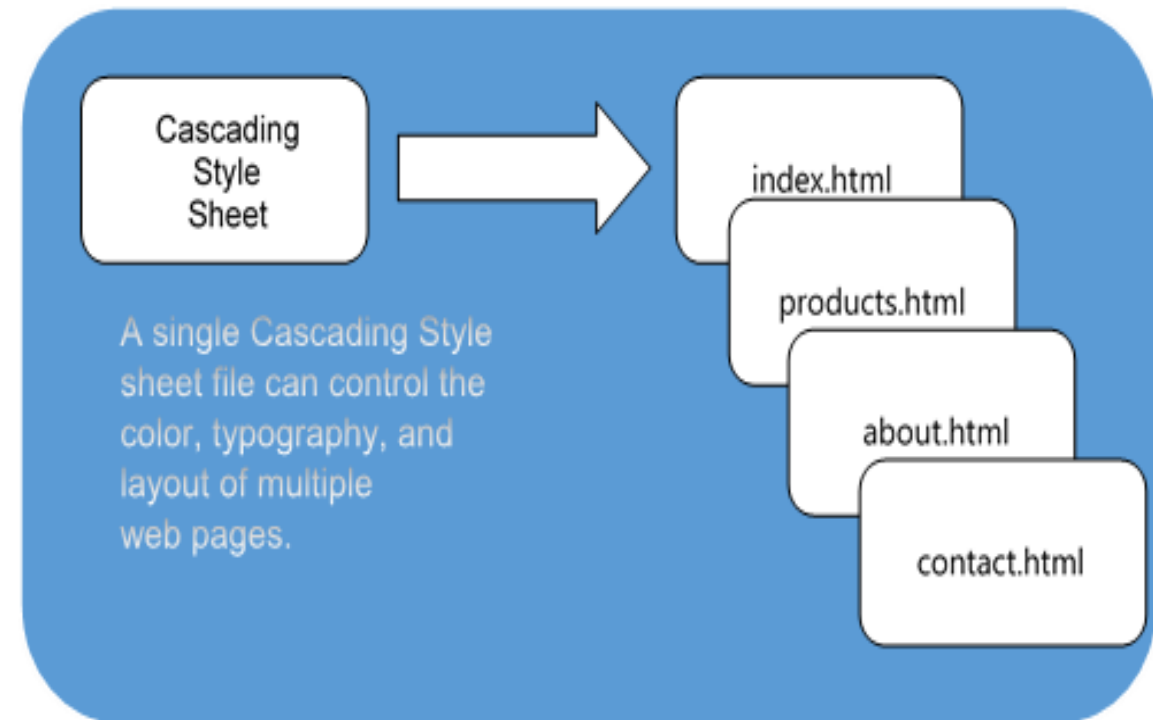


Figure 1: The power of single CSS file

Configuring Cascading Style Sheets

- **Inline Styles**

- Coded as attributes within HTML tags
- Apply only to the specific element

- **Embedded (Internal) Styles**

- Defined in the `<style>` element in the `<head>`
- Apply to the entire web page

- **External Styles**

- Coded in a separate .css file
- Linked via `<link>` element in the `<head>`

- **Imported Styles**

- Use `@import` to bring external styles into embedded or external styles
- Allows combining multiple style sheets

CSS Selectors and Declarations

- Style sheets are composed of style rules
- Each rule has two parts:
 - Selector: Identifies the HTML element to style
 - Declaration: Specifies the property and its value

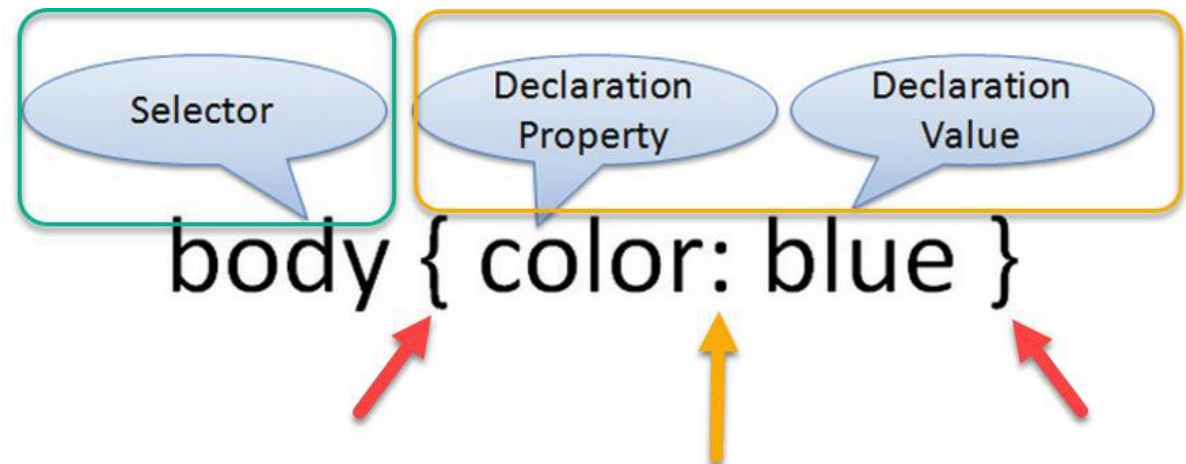


Figure 2: Using CSS to set the text color to blue

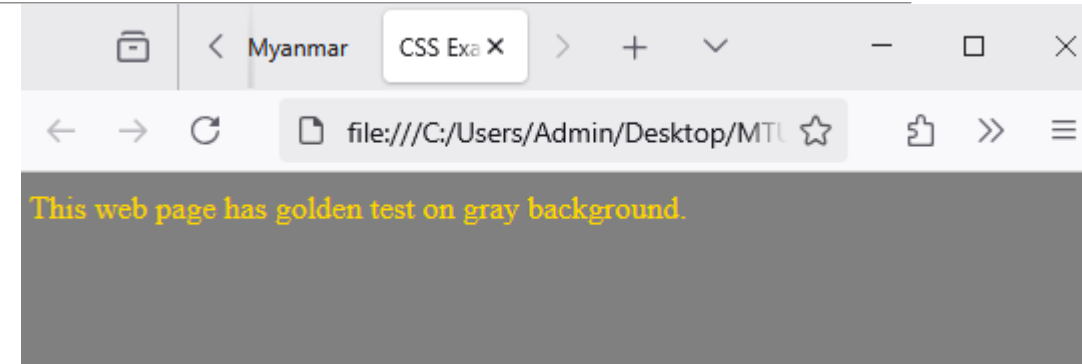
Cont'd

Property	What it does?	CSS style rule
background-color property	configures the background color of an element	<code>body { background-color: yellow }</code>
color property	configures the text (foreground) color of an element	<code>body { color: blue }</code>

Note : To configure more than one property for a selector, use a semicolon (;)
`body { color: gold; background-color: gray; }`

Sample Code

```
<html>
<head>
  <title>CSS Example</title>
  <meta charset="UTF-8">
  <style>
    body{background-color: gray; color: gold;}
  </style>
</head>
<body>
  This web page has golden test on gray background.
</body>
</html>
```



Using Color on Web Pages

- What is RGB Color?
 - Combines different intensities of Red, Green, and Blue
 - Displayed by monitors using RGB values
- RGB Intensity Values
 - Range: 0 to 255 for each color
 - Format: (Red, Green, Blue)
 - Example : `rgb(255,0,0)`
- Hexadecimal Color Values
 - Commonly used for web design
 - Example: `#FF0000` for red, `#00FF00` for green, `#0000FF` for blue

Hexadecimal Color Values

- What is Hexadecimal?
 - Base-16 numbering system
 - Uses characters: 0-9 and A-F
- Hexadecimal in RGB Colors
 - RGB values represented in pairs from 00 to FF (0 to 255)
 - Format: #RRGGBB (e.g., #FF0000 for red, #0000FF for blue)
 - Case Insensitive: #FF0000 and #ff0000 both represent red

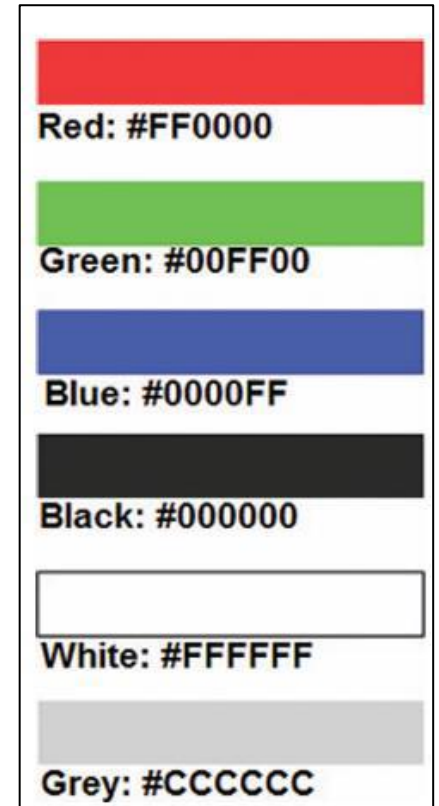


Figure 3: Color swatches and hexadecimal color value

Sample Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>CSS Color Example</title>
  <style>
    h1{color: blue;}
    p{color: rgb(255, 0, 0);}
    b{color: #00FF00;}
  </style>
</head>
```

```
<body>
  <h1>CSS Color Example (Using Color
  Name)</h1>
  <p>This is a red paragraph (Using RGB
  color: rgb(255, 0, 0)).</p>
  This is<b> green</b> text (Using
  Hexadecimal color: #00FF00).</span>
</body>
</html>
```

Inline CSS with the style attribute

- What is Inline Style?

- Coded directly within an HTML tag using the style attribute
- Apply styles to a specific element only

- Structure of Inline Style

- Contains the CSS declaration
- Declaration Format: property: value;
- Multiple properties separated by semicolons (;)

```
<h1 style="color:#cc0000">This is displayed as a red heading</h1>
```

- Example: Single Property

- Example: Multiple Properties

```
<h1 style="color:#cc0000; background-color:#cccccc">  
This is displayed as a red heading on a gray background  
</h1>
```

Sample Code

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>CSS Color Example</title>
    <meta charset="UTF-8">
  </head>
  <body>
    <h1 style="color: #bb00dd;">CSS Inline Example </h1>
    <p style="color: #0099ff; background-color: rgb(122,122,122);">
      This is a blue paragraph on gray background</p>
  </body>
</html>
```

Embedded CSS with Style Element

- What is Embedded Style?
 - Placed within the `<style>` element in the `<head>` section of a web page
 - Used to apply styles across the entire page
- Enclosed between `<style>` and `</style>` tags

Sample Code

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Embedded Styles</title>
    <meta charset="utf-8">
      <style>
        body { background-color: #E6E6FA;
          color: #191970; }
      </style>
    </head>
    <body>
      <h1>Embedded CSS</h1>
      <p>This page uses embedded styles.</p>
    </body>
  </html>
```

Configuring Text with CSS

font-family property	configures font typeface	p { font-family: Arial, Helvetica, sans-serif; }	
font-size property	sets the size of the font		
font-weight property	configures the boldness of the text	font-weight: bold	has a similar effect as the or HTML element
font-style property	configure text displayed in italics	font-style: italic;	has the same visual effect in the browser as an <i> or HTML element
line-height property	modifies the default height of a line of text	line-height: 200%;	

Cont'd

text-align property	configures the alignment of text and inline elements within block display elements such as headings, paragraphs, and divs.	text-align: center;	
text-indent property	configures the indentation of the first line of text within an element.	text-indent: 5em;	The value can be numeric (such as a px, pt, or em unit) or a percentage.
text-decoration property	modify the display of text.	a { text-decoration: none; }	removing the default underline from hyperlinks by setting it to none.

Cont'd

text-transform property	configures the capitalization of text	text-transform: uppercase;	
letter-spacing property	configures the space between text characters.	letter-spacing: 3px;	
word-spacing property	configures the space between words	word-spacing: 2em;	
white-space Property	specifies the way that whitespace (such as a space character, or line feed within code) is displayed by the browser	white-space: normal;	normal (default), nowrap (text will not wrap to the next line), pre (preserves all whitespace in the browser display)

Cont'd

text-shadow property	adds depth and dimension to text	text-shadow: 3px 2px 5px #666;	

Horizontal Offset:

Positive: Shadow on the right.

Negative: Shadow on the left.

Vertical Offset:

Positive: Shadow below.

Negative: Shadow above.

Blur Radius (optional):

0 = sharp shadow.

Higher = more blur.

Color: Any valid color (e.g., #000, red, rgba(0,0,0,0.5)).

Sample Code

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>CSS Font Example</title>
    <style>
      h1 {
        font-family: Arial, Helvetica, sans-serif;
        font-size: 24px;
        font-weight: bold;
        font-style: italic;
        line-height: 300%;
        color: navy; }
```

```
    p {
      font-family: 'Times New Roman', Georgia,
        serif;
      font-size: 18px;
      font-weight: bold;
      font-style: normal;
      line-height: 200%;
      color: darkgreen;
    }
  </style>
</head>
<body>
  <h1>Configuring Text Example 1</h1>
  <p>This is a paragraph demonstrating font-
    family, size, weight, style and line-height.</p>
</body>
</html>
```

Sample Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>CSS Text Properties</title>
  <style>
    .text-transform {
      text-transform: uppercase;
    }
    .letter-spacing {
      letter-spacing: 3px;
    }
    .word-spacing {
      word-spacing: 10px;
    }
    .white-space {
      white-space: pre;
    }
  </style>
</head>
```

```
<body>
  <h1>CSS Text Properties Example</h1>

  <p class="text-transform">
    This text is transformed to uppercase using text-
    transform.
  </p>
  <p class="letter-spacing">
    This text has extra spacing between letters using letter-
    spacing.
  </p>
  <p class="word-spacing">
    This text has extra spacing between words using word-
    spacing.
  </p>
  <p class="white-space">
    This text
    preserves    spaces
    and line
    breaks using white-space.
  </p>
</body>
</html>
```

CSS Selectors

- class selector
- id selector
- descendant selector

class selector

- Apply styles to specific elements on a web page using a CSS class selector.
- Not tied to any particular HTML element.
- Use a dot (.) followed by the class name in CSS.

- In CSS

```
.feature { color: #C70000; }
```

- In HTML,

```
<li class="feature">Usability Studies</li>
```

```
<li class="feature">Search Engine Optimization</li>
```

Sample Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>CSS Class Selector Example</title>
  <style>
    .feature {
      color: #C70000; /* Red color */
    }
  </style>
</head>
<body>
  <h1>CSS Class Selector Example</h1>
  <ul>
    <li class="feature">Usability Studies</li>
    <li class="feature">Search Engine Optimization</li>
  </ul></body></html>
```

id selector

- apply a CSS rule uniquely to a single area on a web page.
- place a hash mark (#) in front of the id name in the style sheet

- In CSS,

```
#feature { color: #5555FF; }
```

- In HTML;

```
<div id="feature">This sentence will be displayed using styles configured in  
the feature id.</div>
```

Sample Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>CSS ID Selector Example</title>
  <style>
    #feature {
      color:#5555FF;
      font-size: 18px;
    }
  </style>
</head>
<body>
  <h1>CSS ID Selector Example</h1>
  <div id="feature">
    This sentence will be displayed using styles configured in the feature ID.
  </div></body></html>
```

descendant selector

- applies styles to elements within a specific container (parent).

- How to use:

```
parent child { property: value; }
```

- Example:

```
main p { color: #00ff00; }
```

Sample Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>CSS Parent-Child Selector
  Example</title>
  <style>
    main p {
      color: #00ff00;
      font-size: 16px;
      font-family: Arial, sans-serif;
    }
  </style>
</head>
<body>
```

```
<body>
  <h1>CSS Parent-Child Selector
  Example</h1>
  <main>
    <p>This paragraph is styled because it is
    inside the &lt;main> element.</p>
    <p>Another styled paragraph within the
    &lt;main> element.</p>
  </main>
  <section>
    <p>This paragraph is not styled because it
    is outside the &lt;main> element.</p>
  </section>
</body>
</html>
```

descendant selector

- defines a section on a web page that is not physically separated from other areas.

This formatting is referred to as inline display.

- Use the `<span>` tag to format an area contained within another tag such as:
 - `<p>` (paragraph)
 - `<blockquote>` (block quotation)
 - `<li>` (list item)
 - `<div>` (division)

```
<p><span class="company">Trillium Media Design</span> will bring
```

Sample Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Span Element Example</title>
  <style>
    .highlight {
      color: red;
      font-weight: bold;
    }
  </style>
</head>
```

```
<body>
  <p>The word <span
class="highlight">important</span> in this
sentence is styled differently.</p>
  <blockquote>
    "This is a <span class="highlight">
key</span> part of the quote."
  </blockquote>
  <ul>
    <li>This item has a <span
class="highlight">special</span> word.</li>
  </ul>
</body>
</html>
```

Using External Style Sheets

- An external style sheet is a text file with a .cssfile extension that contains CSS style rules.
- The external style sheet file is associated with a web page by using the link element.
- This approach provides a way for multiple web pages to be associated with the same external style sheet file.
- The external style sheet file does not contain any HTML tags; it contains only CSS style rules.

link element

- associates an external style sheet with a web page.
- It is placed in the head section of the page.
- a stand-alone, void tag
- Three attributes are used with the link element: **rel**, **href** and **type** (optional in HTML5).

```
<link rel="stylesheet" href="color.css">
```

Sample Code (External CSS design.css)

```
body {  
  font-family: Arial, sans-serif;  
  background-color: #f4f4f4;  
  color: #333;  
}  
header {  
  background-color: #005f56;  
  color: white;  
  text-align: center;  
}
```

```
h1, h2 {  
  font-family: Georgia, 'Times New Roman',  
  Times, serif, sans-serif;  
}  
main {  
  background-color: white;  
}  
footer {  
  background-color: #005f56;  
  color: white;  
  text-align: center;  
}
```

Sample Code (myamar.html)

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Myanmar</title>
    <link rel="stylesheet" href="design.css">
  </head>
  <body>
    <header>
      <h1>Welcome to Myanmar</h1>
    </header>
```

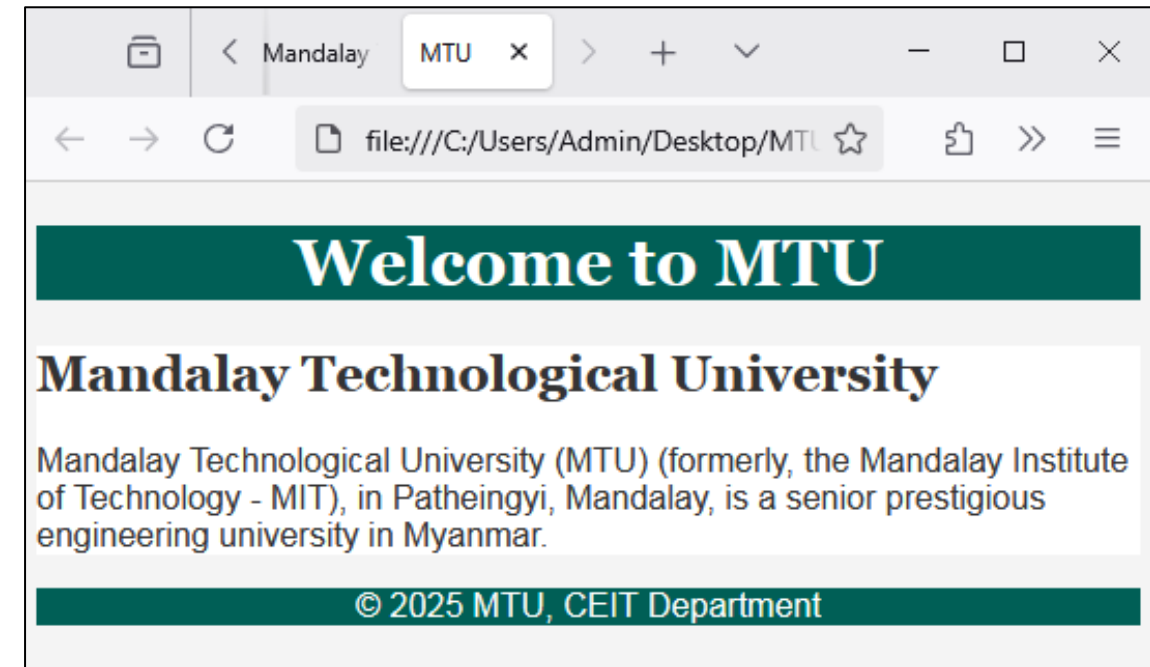
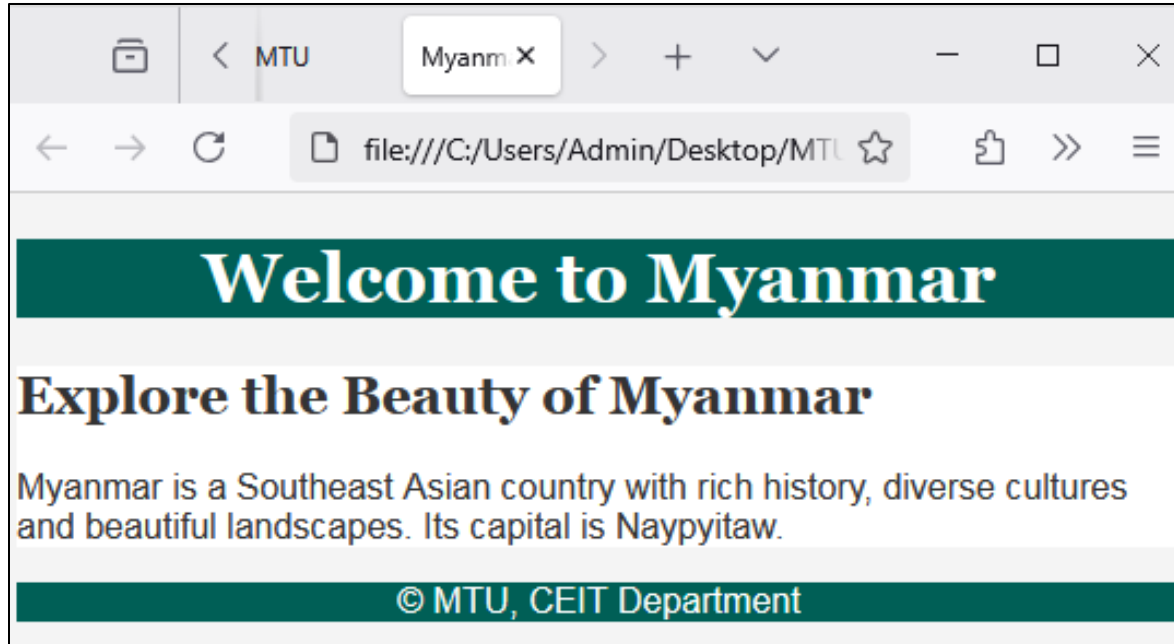
```
<main>
  <h2>Explore the Beauty of Myanmar</h2>
  <p>Myanmar is a Southeast Asian country
  with rich history, diverse cultures and
  beautiful landscapes. Its capital is
  Naypyitaw.</p>
</main>
  <footer>
    <p>&copy; MTU, CEIT Department</p>
  </footer>
</body>
</html>
```

Sample Code (mtu.html)

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Mandalay Technological University</title>
    <link rel="stylesheet" href="design.css">
  </head>
  <body>
    <header>
      <h1>Welcome to MTU</h1>
    </header>
```

```
    <main>
      <h2>Mandalay Technological University </h2>
      <p>Mandalay Technological University (MTU)
      (formerly, the Mandalay Institute of Technology -
      MIT), in Patheingyi, Mandalay, is a senior
      prestigious engineering university in Myanmar.
      </p>
    </main>
    <footer>
      <p>&copy; 2025 MTU, CEIT Department</p>
    </footer>
  </body>
</html>
```

Myanmar and MTU Web Pages



Next Lecture

Visual Elements and Graphics